

**Physical and Biological Characteristics of Brine
Desalination Plants on Marine Ecosystems: Critical Review**

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1. Brief Background

Global population growth and rising consumption continue to put an increasing strain on the world's natural resources. The supply of domestic potable water is one resource that is under severe pressure and is particularly vulnerable to the effects of climate change. As a result, many nations turn to seawater desalination to supplement their other water supply sources. Desalination capacity increased from 5 million m³/day in 1980 to 80.9 million m³/day in 2013 (Ghaffour et al., 2013). In 2016, the water desalination industry grew to around 38 billion m³/year.

Additionally, this volume capacity is expected to grow by approximately 40% by 2030, reaching about 54 billion m³/year (Saleh & Mezher, 2021). Due to the scarcity of potable water in these arid regions and low energy availability, nations in the Middle East were the first to adopt and rely on large-scale desalination (particularly the United Arab Emirates, Kuwait, and Saudi Arabia). Currently, this region produces nearly half of the world's desalinated water ((Saleh & Mezher, 2021). Numerous factors are promoting the expansion of desalination capacity in new parts of the world. Rapid population growth, anticipated changes in precipitation patterns due to climate change, and technological advancements in energy requirements have resulted in many countries with limited water supplies turning to desalination as a secondary source of potable water. The United States, Europe, China, and Australia have expanding desalination capacities (Ghaffour et al., 2013; Saleh & Mezher, 2021). Thus, it is clear that desalination has developed into a critical method for supplying potable water to large cities and industries globally. Desalination plants extract large volumes of seawater and recharge the marine environment with hypersaline brine. Due to the pressing need for water in many parts of the world, marine environmental issues associated with desalination have historically been considered secondary concerns (Safrai & Zask, 2008). Despite this, it is widely believed that desalination plants have a significant potential to harm the physicochemical and ecological characteristics of receiving marine environments (Ariono et al., 2016). With desalination being widely used, it is critical to review and synthesize research on desalination plants' environmental and ecological effects on marine ecosystems. Concerns have centered on the potential impact of hypersaline discharges (hereafter referred to as 'brine') on the salinity of seawater and the effects on marine communities near discharge outlets. However, there is concern about the use and release of toxic antifoulants and antiscalants to maintain plant infrastructure (Ketsetzi et al., 2008), as well as the possibility of thermal stress associated with the discharge of heated effluent from some systems (Lattemann & Höpner, 2008; Roberts et al., 2010; Sadhwani et al., 2005; I. Sola, Fernández-Torquemada, et al., 2020; Tsiourtis, 2001; Zarzo & Prats, 2018). Some activities occur solely during plant construction (e.g., power and sewerage networks) whereas others occur throughout plant lifetime (e.g., pretreatment and post-treatment) ((Galanakis & Agrafioti, 2019; Peters Domènec, 2008). Certain activities (e.g., clearing and grading the project area, connecting to electricity and sewer networks, building facilities and roads, receiving feedwater, discharging by-products, storing and transporting desalinated water, using pretreatment and post-treatment chemicals, generating noise and vibration) alter the natural state of the environment and the (Missimer Robert G., 2018; Wetterau, 2011). Given that desalination plants have a 20 to 35 years lifespan, it is easy to see why environmental factors should be considered as significant as commercial factors (Panagopoulos, 2019). The desalination

plant can also avoid costly economic penalties and operational restrictions by complying with environmental regulations. Thus, the most significant ecological implications are (2) heavy use of chemical products near urban or rural areas, (3) public health risk due to effluents, emissions, or residues (4) landscape alteration ((Dawoud, 2012; Höpner Jens, 1997a; Missimer Robert G., 2018). Brine pollution of the environment Brine is a by-product of desalination that has been studied extensively (Brika, 2019; Gacia Olga; Manzanera, Marta; Ballesteros, Enric; Romero, Javier, 2007; Matsumoto Karen L. M., 2008). . According to the author, desalination facilities create 128,652,000 m³ of brine per day, assuming a 40% freshwater recovery rate. Brine quality and quantity are intimately related to the technology used, feed water quality, standards for produced freshwater, pre- and posttreatment processes, and equipment cleaning and maintenance.

Because brine discharges generate local consequences, even a single plant can have an influence on the marine environment if necessary protections are not taken (Kress, 2019). These problems can be worsened when multiple desalination facilities on a single coast discharge brine. Irreversible dehydration of their cells can cause 'lethal osmotic shock' to marine animals such as fish, plankton, algae, and seagrasses (Abushaban, 2019; Levitt, 2015). Dehydration reduces turgor pressure, which could lead to the long-term extinction of marine animals (Belkin et al., 2018). Seagrass species like *Posidonia oceanica* and *Cynodocea nodosa* have been studied extensively. In *P. oceanica*, salinities above 39.1 mg/L reduce seagrass vitality in terms of leaf growth, necrotic patches, and early senescence. Also, when *P. oceanica* was exposed to 45 mg/L for 2 weeks, 50% of seagrasses died (Fernández-Torquemada & Sánchez-Lizaso, 2005). Similarly, *C. nodosa* (found in the Mediterranean Sea) was negatively affected by increasing salinity in both laboratories and in situ transplantation near brine discharges (Abushaban, 2019; Levitt, 2015). Unlike *C. nodosa* and *P. oceanica* from the Mediterranean, which are sensitive to hypersaline concentrations caused by brine plumes, *Posidonia australis* from Australia has a wide range of salinity tolerances ranging from 27 g/L to 60 g/L. (Cambridge et al., 2019; Garrote-Moreno et al., 2014, 2015; Sandoval-Gil Lázaro; Ruiz, Juan M., 2012). (Cambridge et al., 2017) discovered that *P. australis* could tolerate high salinity for up to 2 weeks at the highest examined salinity (54 g/L) and up to 6 weeks at moderate salinity (46 g/L).

Few studies have investigated brine's impact on benthic species like echinoderms and Polychaeta (Fernández-Torquemada et al., 2013). A recent study found that desalination concentrates can influence benthic microorganisms in a site-specific and localized manner, affecting their number and diversity. (Petersen et al., 2018) found that coral reef-building species have varying salinity tolerances. The study also found that elevated salinity (10% above ambient) changed the coral's physiology and appearance. Heavy marine environments are exacerbated when brine discharges encounter delicate ecosystems. The influence depends on the desalination brine's physicochemical properties and the hydrographic and biological circumstances of the receiving ecosystem. Closed, semi-closed, and shallow areas with lots of marine life are more vulnerable to ecosystem damage. Despite this, it is widely believed that desalination plants have a significant potential to harm the physicochemical and ecological characteristics of receiving marine environments (Ariono et al., 2016). With desalination being widely used, it is critical to review and synthesize research on desalination plants' environmental and ecological effects on marine ecosystems. Concerns have centered on the potential impact of hypersaline discharges (hereafter

referred to as 'brine') on the salinity of seawater and the effects on marine communities near discharge outlets. However, there is concern about the use and release of toxic antifoulants and antiscalants to maintain plant infrastructure (Ketsetzi et al., 2008), as well as the possibility of thermal stress associated with the discharge of heated effluent from some systems (Lattemann & Höpner, 2008; Morton et al., 1997; Roberts et al., 2010; Sadhwani et al., 2005; I. Sola, Fernández-Torquemada, et al., 2020; Tsiourtis, 2001; Zarzo & Prats, 2018). While research has identified several potential mechanisms through which desalination plants may affect marine ecosystems, numerous review articles and case studies published to date cite little or no peer-reviewed literature and provide little or no empirical evidence to support statements about desalination's environmental effects (Areiqat & Mohamed, 2005; Chang, 2015; Malfeito et al., 2005; Sadhwani et al., 2005; I. Sola, Fernández-Torquemada, et al., 2020; I. Sola, Zarzo, et al., 2020). It's unclear whether the potential impacts of desalination plants have been determined through rigorous ecological research or are based on assumptions. The systematic review of peer-reviewed literature was conducted to assess the evidence for environmental effects associated with the discharge of desalination plant effluents. The impact of plant discharges on recipient marine ecosystems' physical properties and biological contaminant are investigated without involving chemical properties of brine. Environmental impact assessments (EIAs) are critical for evaluating the potential consequences of desalination and advocating for appropriate mitigation measures to avoid/minimize these consequences. Numerous strategies can be pursued to improve the desalination process by reducing its environmental impact emphasizing brine discharge.

2. Research Methodology

A comprehensive review of the literature was conducted to record, analyze and evaluate the environmental impacts of desalination and brine treatment. Search engines of databases such as Google Scholar, SCOPUS, and Science Direct were used. Keywords such as “desalination environmental impacts”, “desalination environmental concerns”, “brine discharges”, “marine pollution desalination plants”, “brine treatment environmental impacts”, “desalination plant environmental impact assessment”, “desalination life-cycle assessment”, “desalination environmental monitoring plans” were selected as search terms. Research on emerging desalination and brine treatment technologies, as well as systems/methodologies to minimize energy consumption and air pollution have also attracted substantial attention. In addition, research studies on the marine environment and resource recovery were reviewed due to the complexity of the research topic. 783 publications were found in the primary search attempts (323 papers in Google Scholar, 245 in SCOPUS, and 215 papers in Science Direct). However, there was a degree of overlap between the three databases as observed. When duplicates were deleted, 304 papers remained. Papers were excluded if they were (i) published in a language different from English and (ii) published before 25 years. After this filtering work, 175 papers from 1997 to 2020 remained for the analysis. All search results were read to their full extent, and papers addressing the effects of desalination plants on any aspect of the marine environment were included in the review. Additionally, the reference lists of selected articles were searched to include articles that were not indexed in the three databases or works published prior to the period covered by the three databases. The aspects of the marine environment studied (e.g. salinity, organism abundance, and diversity) were extracted from the journal-title, and the research

methodology used for each article (e.g. monitoring, laboratory experiment, field experiment). When this information was presented, the plant's capacity (in terms of daily discharge), the salinity and temperature of its effluent, and any observed ecological or toxicological effects was recorded. The key results were gathered and analyzed from each research study/review, while the problems and recommendations for future research were also established by highlighting the limitations of the selected research studies.

3. Result and discussion

The literature searches turned up 83 peer-reviewed studies on the environmental and/or ecological implications of desalination plant emissions in receiving marine waters. These journal papers distributed again into based physical characteristics of brine (salinity and temperature) with 16% and with species community 25%. Only four journal papers incorporated manipulative ecological field and case studies. The four different Case studies of 4 different desalination plants: Sydney desalination plant (Australia), Javea SWRO plant (Spain), Alicante Desalination Plant (Spain) and Bousfer Desalination Plant (Algeria) were only taken in this paper. Review, discussion, and opinion pieces made up the majority of reviewed articles (43.5%). While these papers were common, most lacked quantitative evidence and discussed potential effects qualitatively or inductively.

3.1. Physical parameters of Brine

The majority of environmental research on desalination facilities has focused on brines' influence on receiving environments' physical properties parameters such as temperature, pH, electrical conductivity, salinity, total dissolved solids and dissolved oxygen. Only focus is on temperature and salinity on the case studies of 4 different desalination plants: Sydney desalination plant (Australia), Javea SWRO plant (Spain), Alicante Desalination Plant (Spain) and Bousfer Desalination Plant (Algeria). The quality of the feed water, pre-treatment, desalination, water recovery rate, and disposal strategy affect the brine quality and quantity (Table 1). Cleaning, bio-fouling, scale, and foam control chemicals; coagulants used to remove suspended materials; and contaminants resulting from corrosion, such as heavy metals (Lin et al., 2013). Its pre-treatment stages include antiscalants and coagulants that may affect the pH of the water, resulting in brine that has the same temperature as ambient water but has a substantially higher salinity (Missimer Robert G., 2018).

Table: Effects Of Desalination Brines In Marine Ecosystems

<u>Desalination Plant (Location)</u>	<u>Capacity</u>	<u>Brine Salinity (psu) and temperature</u>	<u>Brine discharge and extension</u>	<u>Species community</u>	<u>Summary of finding</u>	<u>Reference</u>
Sydney Desalination Plant New South Wales, Australia,	250 ML per day of drinking water	The hypersaline effluent averages 20.1 (2.0) C and 59.6 (5.7) PSU conductivity. The ambient seawater temperature around the exit ranges from 14.5 to 24°C.	brine that is discharged into the ocean from an outlet around 300 m offshore and ~ 25 m deep	impact to <i>Ecklonia</i> , <i>radiata</i> , <i>kelp</i> and algal turfs persisted	Six-year M-BACI assessment of desalination discharge outlet construction and hypersaline effluent consequences on temperate reef habitat-forming species cover. The desalination outlet construction reduced the cover of <i>Ecklonia radiata</i> (kelp) and increased the cover of algal turfs up to 55 m away. After the hypersaline brine discharge began, the impact on <i>E. radiata</i> and algal turfs persisted, but was reduced to less than 25 m from the outflow.	(Kelaher & Coleman, 2022)
Javea SWRO plant, Spanish Mediterranean coast	produce 6720 m3 of freshwater per day; thus, its maximum freshwater production capacity is 26,800 m3 per day with a recovery rate of 44.8%	Salinity: diluted BD from the Javea SWRO plant was shown to have an average salinity of <44 PSU Temperature: 20.1 ± 1.1 °C)	during summer, the saline plume area could extend to a distance of about 300 m from the channel outfall	<i>Posidonia oceanica</i> <i>meadow</i> <i>seagrass</i> <i>Echinoderm</i> . Fish larvae	The results of 17 years of BD monitoring indicate how scientists and industry may work together to reduce environmental impacts. The brine was diluted before being discharged into a low-value artificial waterway. Shoot production and echinoderm abundances were similar in control and impact areas across time series. The BD area had a higher abundance and species richness of fishes. The key findings should be considered for future initiatives.	(Sola et al., 2020)
Alicante I desalination plant (Spain)	50,000 m3/day using reverse osmosis technology (RO) with a recovery of 42–44%.	salinity threshold of 38.5 PSU Temperature: 14–30°C.	2000 m away from the coastline	<i>Posidonia oceanica</i>	salinity thresholds in Spanish desalination plants' EIS. In 2005, the EIS imposed salinity standards lower than those advised by scientists to conserve <i>Posidonia oceanica</i> meadows, but after that, thresholds more in line with scientific recommendations were created. The energy consumption of these lower thresholds was also studied using the Alicante desalination facilities as a case study.	(Navarro Barrio et al., 2021)
Bousfer" desalination plant, (ALGERIA)	reverse osmosis plant that treats daily 12480 m3 of sea water to provide 5500 m3 of fresh water with a conversion rate of 45%.	Salinity :35- 45 psu Temperature 13-16°C .	located 250 m west and 150m from east	<i>Patella rustica</i> limpets	The consequences of these brine discharges on a marine gastropod mollusk, <i>Patella rustica</i> , were studied. This was done by measuring defence and damage biomarkers in this species' tissues. <i>P. rustica</i> limpets were sampled and brine and surface coastal waters were analysed at four sites (H, W, E and R). This ecotoxicological biomonitoring revealed the harmful impact of desalination plant brine discharges in this cold season. Environmental disturbances had a greater impact on the limpets obtained from site H. Our ecotoxicological analysis showed that the winter discharges from the Bousfer desalination plant are detrimental, despite the fact that production is reduced and the plant is shut down for maintenance and cleaning.	(Benaissa et al., 2020)

3.1.1 Salinity

Lots of desalination research has been done on brine plumes in receiving waters. A published study shows desalination plant effects on receiving water salinity vary (Table 1). There have been reports of salinity plumes extending hundreds of metres or even kilometres (Benaissa et al., 2020; Kelaher & Coleman, 2022; Navarro Barrio et al., 2021; F. Sola et al., 2020). The variability in these results is likely attributable to a combination of plant capacity, diffuser design, environmental hydrology ((Cambridge et al., 2019; Höpner Jens, 1997b) Ho and study sampling effort (i.e. their power to detect changes, which is dependent on the amount of sampling and

sampling design). Within 20m of the exit, the plume normally fades to less than 2 ppt above the background salinity (Table 1). Hundreds of meter-long plumes were usually only slightly above background levels, usually less than 0.5 psu. Notably, most of this research is on desalination plants in the Mediterranean Sea (Table 1). Brine plumes tend to extend deeper below the seafloor than natural saltwater plumes (Gacia, 2007; Sanchez-Lizaso et al., 2008). This is important biologically because benthic organisms are more susceptible to brine discharges than pelagic and planktonic organisms. For example, porewaters reveal brine discharges to seagrass meadows that overlying fluids do not. To model this brine exposure situation, seagrass has been exposed to vertically stratified salinities in the laboratory. Regardless of exposure type, seagrass survival was significantly reduced.

3.1.2 Temperature

Some desalination plants also raise the temperature of the brine relative to the receiving waters, albeit this is covered in significantly fewer studies in the current study. Several writers have proposed that high temperatures in receiving waters may have influenced desalination plant biological consequences. It has been shown that multi-stage flash distillation (MSF) can produce brines 10–15°C warmer than marine input waters (Höpner Jens, 1997a; Lattemann & Höpner, 2008). Reverse osmosis is becoming more popular, resulting in temperature plumes. Current Studies on thermal effects of brines have yielded mixed results, similar to those on salinity effects. For example, in Western Australia, a multi-purpose power and desalination plant discharge might raise receiving water temperatures by average 18 °C in a 7 square kilometer area surrounding outfalls (Kelaher & Coleman, 2022) Other investigations reported minor thermal impacts near outfalls despite desalination, raising intake water temperatures by up to 35 °C (Navarro Barrio et al., 2021).

3.2 Biological Communities

Desalination brine discharges have severely impacted *Posidonia oceanica* seagrass for a prolonged period (Benaissa et al., 2020; Kelaher & Coleman, 2022; Navarro Barrio et al., 2021; F. Sola et al., 2020). Seagrass plays a crucial role in the marine ecosystem; meadows of *Posidonia oceanica* can shelter associated algae, invertebrates and vertebrates in areas of high biodiversity, and contribute to improving water quality, preventing coastal erosion, and regulating biochemical fluxes along the coast. *Thalassia testudinum* was unaffected by a salinity increase of up to 4 ppt from an RO desalination discharge, and both *Cymodocea nodosa* seagrass and *Caulerpa prolifera* green algae were unaffected by a discharge plume as long as the salinity remained below 37.5 ppt, which is within their natural salinity range (Gacia et al., 2007); whereas a 1–2 ppt increase above ambient salinity of 37–38 ppt caused significant effects on *P. oceanica*. *P. oceanica* is sensitive to increased salinity due to brine discharge and was substantially impacted. The plant's tissues suffered an increase in nitrogen content and reduced glutamine synthetase activity and its health was impaired, particularly in the area next to the discharge where the salinity is higher, the deleterious impacts on the plant's health being because of a higher occurrence of necrosis, a reduction in total non-structural carbohydrate content, and reduced leaf growth (Gacia et al., 2007; Sánchez-Lizaso et al., 2008). The structure and vitality of *P. oceanica* were severely affected at a salinity of 39.1 ppt and 38.4 ppt, respectively, and the plant's mortality considerably increased above 40 ppt, as 50% of *P. oceanica* died upon the

exposure to 45 ppt salinity within 15 days (Sánchez-Lizaso et al., 2008). Increased salinity can harm seagrass by decreasing root development, increasing osmoregulators such as amino acids and sugar, and decreasing potassium and calcium ion concentrations (Cambridge et al., 2017). Desalination brine discharge increases nutrient contents, including nitrates, leading to increased epiphyte abundance (Gacia et al., 2007). But in Sydney desalination plant, there was high impact environmental caused a decrease in the cover of *Ecklonia radiata* (kelp) and an increase in the cover of algal turfs up to 55 m from the outlet (Kelaheer & Coleman, 2022).

3.3. Recent Case Studies:

3.3.1 Sydney Desalination Plant

The influence of a large-scale desalination plant's discharge outlet and hypersaline effluent on habitat-forming reef species and the extent of bare rock and sand patches was investigated. The Sydney Desalination Plant was the subject of our study. This facility, which uses reverse osmosis to produce up to 250 ML of drinking water per day (Kelaheer et al., in review), can meet about 15% of Sydney's freshwater needs (Kelaheer et al., 2020). When fully operational, the Sydney Desalination Plant produces 342 (32) ML of hypersaline brine per day, which is discharged into the ocean from a 300 m offshore and 25 m deep outlet (Kelaheer et al., 2020). The average (SD) temperature of the hypersaline effluent is 20.1 (2.0) °C, with a conductivity of 59.6 (5.7) PSU (Kelaheer et al., in review). The ambient seawater temperature at the exit varies between 14.5 and 24.0 °C, with an average salinity of around 35.4 PSU (Clark et al., 2018). (Roughan and Morris, 2011). the findings of a six-year M-BACI study on the effects of brines cover of habitat-forming species on temperate reefs. The installation of the desalination outlet resulted in a decrease in *Ecklonia radiata* (kelp) cover and an increase in algal turf cover up to 55 meters from the outlet. The impact on *E. radiata* and algal turfs persisted after the discharge of hypersaline brine began, but it was reduced in geographical extent to less than 25 m from the outflow.

3.3.2 Alicante Desalination Plant

The study looks at how salinity thresholds are used in Environmental Impact Statements (EIS) for desalination plants in Spain. In the EIS of 2005, lower salinity limits were established to protect the *Posidonia oceanica* meadows than those proposed by the scientific community, however, after that year, thresholds more commensurate with scientific recommendations were created. Furthermore, the energy consumption of these lower thresholds was investigated using the brine discharge of the Alicante desalination plants as a case study. The 38.3 psu threshold was easily exceeded, resulting in a higher dilution ratio and energy consumption. For six months, the dilution cost more than 600,000 kWh and €50,000. If the implemented value were 38.5 PSU, the dilution ratio and energy consumption would have been significantly reduced with no detrimental environmental effects on the marine ecosystem. To avoid additional energy-economic costs, it is recommended that protection of *P. oceanica* meadows salinity threshold Spanish regulation can be unified based on the best available scientific evidence.

3.3.3. Javea Desalination Plant

The most severe problem about desalination's long-term viability is its environmental impact. The findings of a long-term environmental strategy to monitor brine discharge (BD) from a desalination plant in a high-environmental-value area in Spain are presented. The biological

characteristics of biological communities were studied using generalized additive models. The results of 17 years of BD monitoring show how well-planned decision-making between scientists and industry can reduce the company's environmental impact. The diluting of brine prior to discharge into a low-value artificial channel greatly reduced the brine influence region. *P. oceanica* shoot production and echinoderm abundances were largely steady throughout time, with similar values seen in control and impact areas. On the other hand, the BD area has a larger abundance and diversity of fish species. The essential insights presented here should be taken into account for future use in similar projects.

3.3.4 Bousfer Desalination Plant

The research paper aims to determine the impact of brine discharged on the Algerian west coast by the Bousfer desalination plant during the winter months (February 2016), when output is decreased to 2500 m³/day and maintenance operations are temporarily halted. Sublethal and chronic effects on the marine gastropod mollusk *Patella Rustica* exposed to brine discharges were investigated in this study. This was accomplished by measuring biomarkers of defence and injury in the tissues of this species. At four different locations, *P. Rustica* limpets were sampled and Physico-chemical analyses of the brine and surface coastal waters were performed (H, W, E, and R). This ecotoxicological biomonitoring revealed the desalination plant's brine discharges had a harmful influence during this cold spell. Indeed, the limpets obtained from site H were more affected by all environmental perturbations than those gathered from the other sites. In the population of site H, antioxidant defence and biotransformation enzyme reactions were at their peak. This oxidative stress also has a genotoxic effect, as evidenced by ADN and lipid damage. Even though output is reduced and the plant even observes brief stoppages for maintenance and cleaning operations throughout the winter, our ecotoxicological diagnosis revealed the adverse effect of Bousfer desalination plant discharges.

4. Conclusion

An examination of the literature reveals a broad belief and acceptance that desalination plants constitute a major danger to marine ecosystems. The evidence for desalination brines' salinity, temperature, and pollutant effects on receiving water quality are fairly apparent; nevertheless, when brines are delivered into well-flushed settings, the effects are typically minor (10 s of meters). When brines and their constituents are present in adequate amounts, laboratory trials, toxicological examinations, and manipulative field experiments all reveal that they have the ability to cause harmful effects on aquatic life. In some situations, significant toxicological impacts of desalination brine on marine vertebrates and invertebrates have been discovered at dilutions typical of those encountered near desalination outfalls. As a result of our analysis of the literature, we conclude that desalination plants may have a detrimental effect on the ecology of marine environments. While some earlier studies discovered broad-scale effects on the ecology of receiving environments, recent research indicates that appropriate discharge site selection, ocean current modelling, and proper plant maintenance and operation will all help to limit the spatial extent of desalination plant discharges' ecological effects.

One area where there is a distinct absence of evidence is in field-based ecological monitoring. Unfortunately, many of the ecological monitoring programs that have been reported do not appear to be academically acceptable effect evaluations. Thus, there is a widespread dearth of scientific evidence to support conclusions about the consequences of desalination brines in receiving systems, a reality recognized in nearly every region with big plants. Only in seagrass environments have biological monitoring studies been integrated with laboratory and field experiments to examine the impacts of brines on seagrass ecosystems (Sánchez-Lizaso et al., 2008). Additionally, professional experience indicates that a substantial amount of study regarding the biological and environmental effects of desalination facilities is likely to exist in the grey literature (i.e., unpublished technical reports produced by consultants and government bodies). Access to this literature is notoriously difficult for the purpose of conducting a literature review. It is critical to promote and encourage scientists conducting such research to publish their findings in peer-reviewed publications in order to increase knowledge in this field. Effective monitoring procedures can quantify the spatial extent of desalination discharge impacts and are essential to inform future site selection criteria for discharges.

It is worth noting that many published manuscripts purport to describe or review the ecological impacts of desalination plants, but cite little or no peer-reviewed literature, provide scant or no details of the methodology and statistical analysis. Environmental research must transition from qualitative to quantitative methodologies, using strong experimental designs similar to those used in other fields of marine research to quantify ecological implications. With the expansion of desalination capacity occurring in several places throughout the world, there is an unavoidable necessity to monitor their effects on marine ecosystems using good and acceptable scientific methods.

The following major areas for future research.

1. Field manipulation experiments to study desalination brine impacts. In this review, only three papers used manipulative experiments in the field (Benaissa et al., 2020; Kelaher & Coleman, 2022; F. Sola et al., 2020). But these studies were able to observe impacts on multiple species simultaneously, show that small salinity shifts (1–2 ppt) can have large impacts on exposed communities, and provide insight into the brine constituents responsible for observed effects. These types of field tests can be difficult to execute, but the results are plainly invaluable. These studies could replicate effluent release at various flow rates to assess repercussions at various exposure levels.

2. Before–After Control-Impact (BACI) procedures use several reference locations and repeated sampling before and after plant operation. Ecological monitoring strategies should contain several reference locations and duplicated sampling before and after the relevant activity. The possible ecological implications of desalination plants have been studied in Australia, however the results have not yet been published in peer-reviewed journals (Kelaher & Coleman, 2022). It is difficult to demonstrate an influence without adequate reference locations and baseline estimations of ecological state, which is troublesome for both operators and regulators. These designs often necessitate sampling at least five reference locations three times before and during plant operation (annually where possible to avoid confounding by seasonal changes). The confidence in post-operation studies is weaker since any variations between reference and potentially impacted areas may have existed prior to the plant's operation (García-Menéndez et al.,

2021; Höpner & Windelberg, 1997; Lattemann Thomas, 2008)). In the absence of pre-construction data, a research with several sampling times and independent reference locations should offer a realistic estimate of an existing plant's effects. It has been suggested that a variety of statistical models and philosophical approaches to analysis be reviewed as part of any new ecological monitoring programme design

3. Interactions, synergistic and additive effects of a range of meteorological variables and desalination brines. Climate variations may have additive or synergistic effects on marine community reactions to desalination brines. Toxicology testing could be undertaken utilising local organisms from the developing location under a variety of environmental conditions (e.g. temperatures). Ecological monitoring studies could potentially investigate the possibility for seasonal differences in marine community responses to desalination brines.

4. Specific tests of desalination plant anti-scalants. There is a lack of fundamental toxicological information provided on routinely used anti-scalants found in brine effluent. Studies to optimise anti-scalant use are needed to reduce brine inclusion.

5. Publish studies in scientific journals. Much of the research on desalination facilities' consequences has been published in the grey literature. These research should be published in peer-reviewed journals to help define desalination plant design, location, and management.

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